Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/05/2018
PI Name:	Nestorovic, Ned M.S.		
Project Title:	Development of a Proof-of-Concept in-Flight Unobtrusive Retinal Imaging System		
Division Name:	Human Research		
Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Smart Medical Systems and Technology Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	98072	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI-RFA-SMARTCAP
Start Date:	09/01/2016	**End Date:**	05/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NCC 9-58-SMST00016		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Prolonged exposure to microgravity results in multiple alterations to the visual system. Currently the equipment on the International Space Station (ISS) to view/monitor the retina is too large, requires two astronauts to operate, requires pupil dilation, and is prone to breaking down. Our team has developed a proof-of-concept retinal imaging device that will only require one astronaut to get a retinal image (self-imaging) and astronauts will not have to have their pupils dilated. This will save significant astronaut time and minimize crew training time due to its ease of use. In June of 2017, our team demonstrated the proof-of-concept retinal imaging device for a team of NASA eye doctors and key staff working on the Vision Impairment and Intracranial Pressure (VIIP) issue. The NASA group indicated that they were very satisfied with the results of the demonstration and agreed that the device does prove that a person (potentially an astronaut) can image their own retina without pupil dilatation. Additionally, the team demonstrated supporting software that showed, in real time, the retinal image on a laptop in high-resolution and has a 40+ degree wide view of the retina. Additionally, the retinal image was able to be magnified and maintain a high resolution image. Using this proof-of-concept device and the information derived from it, the team is planning to continue development of the device (Phase 2). The team believes it can reduce the size of the device to a point where it could be worn as a headset, similar in size and design of a virtual reality headset. This will make the device much easier for astronauts to set up and use, saving time and astronaut training time. The system will be a fully functional retinal imaging device, that can be used on adult eyes with focus accommodation on the high-resolution sensor and automatic image capture. The system will show real-time video acquisition, using a laptop or tablet. It will also demonstrate cropping, resizing, and basic stitching of two overlapping images if available. This cropping, resizing, and basic stitching will eventually be done manually/automatically and will have the ability to automatically compare a base-line image to the current image and indicate where/if there are differences/changes in the retina.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This technology has many potential benefits and applications in the commercial market on Earth. The retinal imaging device could be developed to be a: - Low cost at-home monitoring device that will provide eye doctors with vital and timely images of patient's retinas without the need to come into the office; - Part of a kiosk health monitoring system that can provide not only a comprehensive tele-medicine eye exam but an overall health assessment; - Low-cost retina screening device in remote areas where minimally skilled health workers can provide a retinal image for doctors via a tele-medicine format.
Task Progress:	Prolonged exposure to microgravity results in multiple alterations to the visual system. Currently the equipment on the ISS to view/monitor the retina is too large, requires two astronauts to operate, requires dilation, and is prone to breaking down. As a result, images of the retina are taken very infrequently at a rate of about once of month. This needs to be replaced with a more space-friendly, unobtrusive device where imaging can occur in a matter of minutes and doesn't require the support of another astronaut. In doing so, retinal images can be acquired more frequently. We developed the proof-of-concept retinal imaging device device to overcome the issues stated above with the current retinal imaging equipment and successfully demonstrated it to a NASA team of VIIP experts at Johnson Space Center (JSC) in June, 2017. The primary deliverables of the proof-of-concept demonstrator included the following: A very high resolution imaging system and retinal illumination system. Some benefits of this technology, compared to the existing retinal imaging systems on the ISS are: • single astronaut user; • non-mydriatic; • under three-minute reading of both eyes; • up to 40 degrees of high resolution retinal image • reasonably small form factor and no requirement for cabin dimming or other environmental changes The team also created and demonstrated accompanying software that provided a real-time image of the retina being imaged on a nearby laptop screen. This software gave a high-resolution image that was 40+ degrees wide (without dilating the pupil) and was able to be magnified (and maintain the high-resolution image) to provide a close up view of various sections of the retina.
Bibliography Type:	Description: (Last Updated:)